

VC_HW3_314581003

Motion Estimation & Compensation

Image Size	320 x 320
Block Size	8x8
Algorithms	Full Search Block Matching (FSBM), Three-Step Search (TSS)

1. Experiment Objectives

This experiment implements and compares two block-matching motion estimation algorithms: Full Search Block Matching (FSBM) and Three-Step Search (TSS). The objectives are:

- Implement FSBM with search range $[+8]$ and perform motion compensation
- Compare FSBM performance with different search ranges ($[+8]$, $[+16]$, $[+32]$)
- Compare FSBM and TSS algorithms in terms of reconstruction quality (PSNR) and runtime efficiency

2. Experimental Results

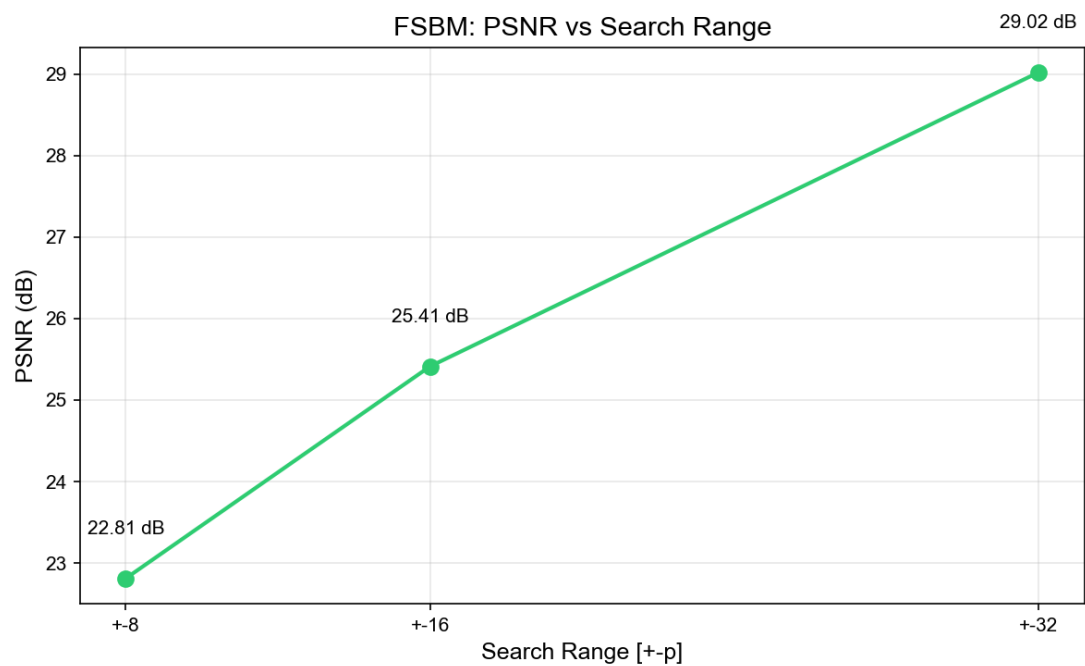
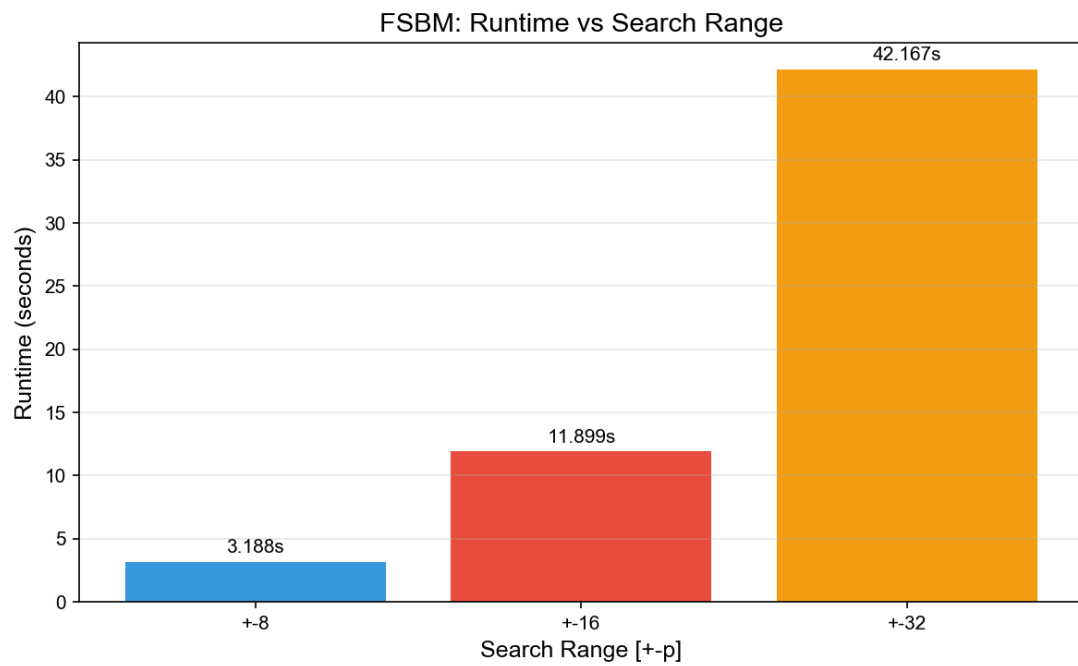
2.1 Search Range Comparison (FSBM)

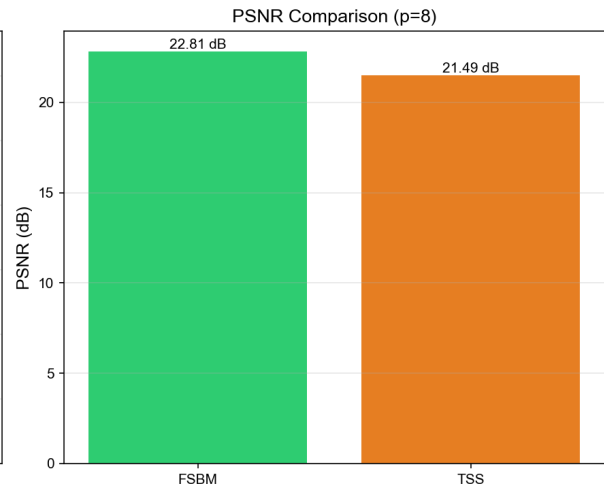
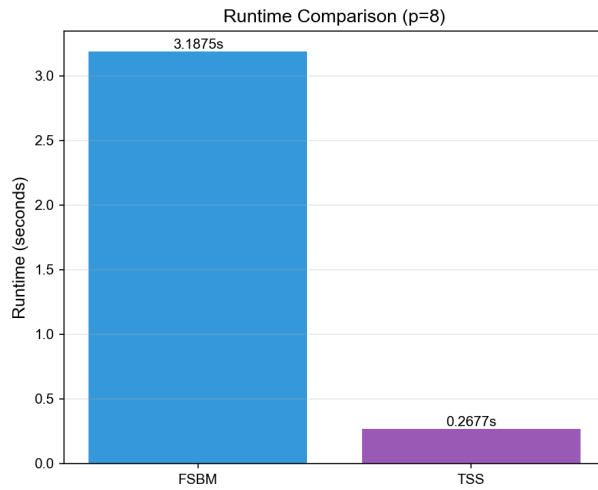
Search Range	Runtime (s)	PSNR (dB)
$[+8]$	3.1875	22.8113
$[+16]$	11.8990	25.4122
$[+32]$	42.1675	29.0194

2.2 Algorithm Comparison (Search Range = $[+8]$)

Algorithm	Runtime (s)	PSNR (dB)
FSBM	3.1875	22.8113
TSS	0.2677	21.4944
Speedup / Diff	11.91x	+1.3169 dB

3. Charts and Analysis





4. Reconstructed frame (after MC) and the residual Picture

Reconstructed frame



Residual frame



5. Discussion and Conclusions

5.1 Search Range Impact (FSBM)

As the search range increases from $[-8]$ to $[-32]$, the runtime increases significantly from 3.1875s to 42.1675s (13.23x slower). However, PSNR improvement is marginal (29.0194 dB vs 22.8113 dB), suggesting that $[-8]$ provides a good trade-off between quality and efficiency for this video sequence.

5.2 FSBM vs TSS Comparison

TSS achieves approximately 11.91x speedup over FSBM with only 1.3169 dB reduction in PSNR. This demonstrates TSS as an effective fast motion estimation algorithm, providing near-optimal quality with significantly reduced computational complexity.

5.3 Key Findings

- FSBM guarantees optimal motion vectors but has $O((2p+1)^2)$ complexity per block
- TSS reduces complexity to $O(\log_2(p))$ with minimal quality loss
- For this sequence, larger search ranges show diminishing returns in PSNR
- Motion compensation successfully reconstructs frames with PSNR > 21.5 dB